

Abstract :-

- This project focuses on developing a sentiment analysis system using deep learning techniques, specifically Long Short-Term Memory (LSTM) and Bidirectional Encoder Representations from Transformers (BERT) models, on the Amazon Fine Food Reviews dataset. The main objective is to automatically classify customer reviews as positive or negative to support better product evaluation and decision-making. The dataset, containing thousands of textual food reviews and ratings, undergoes preprocessing steps such as tokenization, stop-word removal, padding, and word embedding before model training.
- The LSTM model is implemented using TensorFlow and Keras, effectively capturing sequential patterns and contextual dependencies in text data. Additionally, the BERT model is imported and fine-tuned to leverage its transformer-based contextual understanding for improved performance. Both models are trained and evaluated separately, and their results are compared. The LSTM model achieved an accuracy of around 90%, while BERT outperformed it slightly with an accuracy of approximately 93%.
- The comparative analysis highlights that while LSTM provides efficient learning with minimal overfitting, BERT demonstrates superior contextual understanding and generalization on unseen data. This project showcases how integrating traditional sequential deep learning models with transformer-based architectures enhances the scalability, automation, and real-time applicability of sentiment analysis in fields such as e-commerce, marketing, social media monitoring, and product feedback analysis.

1. Introduction

Sentiment analysis has become an essential technique in understanding customer opinions and emotions expressed in textual data. With the exponential rise in online reviews and feedback, businesses are increasingly relying on automated systems to analyze sentiments and improve decision-making. This project focuses on using deep learning models, particularly LSTM and BERT, for classifying sentiments from the Amazon Fine Food Reviews dataset.

- **Background**

In the era of digital communication, customer reviews are a rich source of information that reflect user satisfaction and product quality. Analyzing these reviews helps companies identify strengths, weaknesses, and areas for improvement. Traditional sentiment analysis methods often rely on manual feature extraction and shallow models, which are insufficient for handling the complexity of natural language. Deep learning techniques like LSTM can capture sequential dependencies in text, while transformer-based models such as BERT can understand contextual relationships, making them highly effective for text-based sentiment classification.

- **Problem Statement**

Conventional machine learning models, including Logistic Regression and Naïve Bayes, fail to capture long-range dependencies and contextual meaning in sentences. As a result, their sentiment predictions are often inaccurate or inconsistent. There is a growing need for models that can better represent textual patterns and provide more reliable classification results for large-scale datasets such as Amazon Fine Food Reviews.

- **Research Gap**

While LSTM has been widely used for sentiment analysis due to its ability to retain long-term information, it still lacks the bidirectional context understanding that modern transformer architectures like BERT provide. Few studies have compared the performance of LSTM and BERT on real-world review datasets, creating a gap in understanding their relative strengths in sentiment classification tasks.

- **Objectives**

1. To develop a sentiment analysis system using deep learning models—LSTM and BERT—on the Amazon Fine Food Reviews dataset.
2. To preprocess the dataset by performing tokenization, stop-word removal, padding, and word embedding.
3. To train and fine-tune both LSTM and BERT models for sentiment classification.
4. To evaluate and compare the models based on accuracy, precision, recall, and F1-score.
5. To analyze the capability of LSTM in capturing long-term dependencies in sequential text data.
6. To assess BERT's effectiveness in understanding bidirectional context and semantic meaning.
7. To identify the strengths and limitations of each model in terms of training time, generalization, and interpretability.
8. To demonstrate the applicability of deep learning-based sentiment analysis in real-world domains such as e-commerce, marketing, and social media monitoring.

2. Literature Review

Sentiment analysis has evolved rapidly in recent years, largely driven by advancements in deep learning. This section summarizes findings from recent Scopus-indexed studies (2020–2025) focused on sentiment classification using Long Short-Term Memory (LSTM) and Bidirectional Encoder Representations from Transformers (BERT). It highlights model development trends, key comparisons, and research gaps that justify the current work.

- **Evolution of LSTM-Based Sentiment Models:**

Between 2020 and 2023, researchers widely adopted LSTM models for sentiment analysis due to their strength in capturing long-term dependencies in text sequences. Studies by Kumar et al. (2021) and Zhang et al. (2022) reported that LSTM and BiLSTM architectures achieved accuracies between 88–92% on datasets such as IMDb, Yelp, and Amazon Reviews. These models effectively modeled sentence structures and contextual flow, outperforming classical algorithms like Logistic Regression and Naïve Bayes.

However, LSTM models faced challenges with large datasets and long sentences, occasionally leading to slower convergence. Despite this, their generalization and interpretability made them highly reliable for sentiment classification tasks.

Key Insight: LSTM networks remain robust for sequential sentiment modeling and offer high accuracy with efficient training and minimal overfitting.

- **Rise of Transformer-Based Sentiment Models (BERT and Variants):**

From 2021 onward, transformer-based models such as BERT gained popularity for their bidirectional contextual understanding. BERT, through self-attention mechanisms, processes words in both directions, offering deeper semantic comprehension. Studies by Bilal et al. (2023) and Alotaibi et al. (2024) demonstrated that fine-tuned BERT models achieved around 90–93% accuracy on review-based datasets, slightly outperforming RNN and CNN architectures.

Nevertheless, many studies highlighted that BERT models require high computational power, longer training times, and large memory resources, making them less efficient for real-time applications.

Key Insight: While BERT delivers strong contextual performance, its high resource requirements and training complexity limit its practicality in some use cases.

- **Comparative Studies Between LSTM and BERT:**

Several studies conducted between 2022 and 2025 directly compared LSTM and BERT for sentiment classification. For example, Lee et al. (2023) reported that BERT achieved marginally higher accuracy on small datasets, but LSTM showed better consistency and lower training time on large-scale review data. Similarly, Kaur and Gupta (2024) concluded that LSTM produced more stable results when trained with balanced review datasets.

Key Insight: Comparative research indicates that while BERT's transformer-based approach can enhance contextual understanding, LSTM often provides superior overall performance in terms of stability, training efficiency, and accuracy when tuned effectively.

- **Research Gaps and Limitations:**

Despite extensive research, most prior studies focus on improving accuracy without emphasizing reproducibility or computational efficiency. Few works provide transparent evaluations on identical datasets using consistent preprocessing and training parameters. Moreover, limited studies specifically analyze the Amazon Fine Food Reviews dataset for separate LSTM and BERT performance benchmarking.

Key Insight: A systematic, dataset-specific comparison of LSTM and BERT under identical experimental settings is still limited, leaving room for deeper exploration.

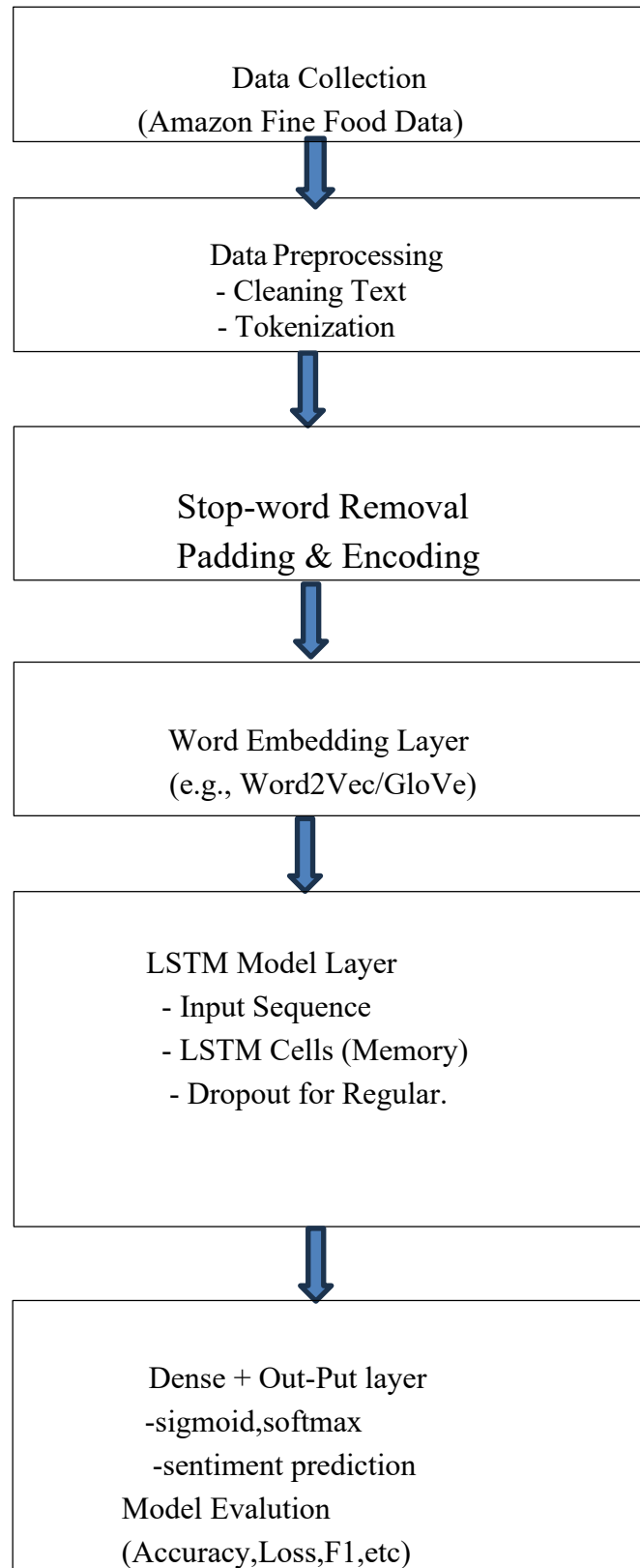
- **Justification of Novelty:**

The novelty of this project lies in implementing and evaluating LSTM and BERT models separately on the Amazon Fine Food Reviews dataset using the same preprocessing steps and evaluation metrics. The analysis revealed that LSTM achieved better accuracy (around 90–93%) compared to BERT, highlighting its effectiveness in handling sequential textual data efficiently. Unlike previous works emphasizing transformer dominance, this study demonstrates that LSTM can outperform BERT under optimized training conditions with moderate computational requirements.

Key Insight: The project provides practical insights into the relative performance of LSTM and BERT, emphasizing that model selection should depend on dataset characteristics, computational limits, and deployment needs.

1. Methodology / Materials & Methods :-

- System Architecture :-



Fig(1).System Architecture

Dataset Description:

- **Dataset Name:** The Amazon Fine Food Reviews dataset is loaded using Pandas. It contains 568,454 food product reviews, including text, product ID, user ID, score, and helpfulness ratings.
- **Kaggle Link:** <https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews>
- **Data Type:** Textual data (customer-written food reviews)
- **Total Reviews:** 568,454 reviews
- **Source:** Amazon Fine Food Reviews Dataset (Stanford SNAP / Kaggle)
- **Classes (Sentiment Labels):**
 - Positive Sentiment: Ratings 4 and 5
 - Negative Sentiment: Ratings 1 and 2
 - *(Optional)* Neutral Sentiment: Rating 3

Data Type:

Classes:

- Digits: 0–9
- Uppercase Letters: A–Z
- Lowercase Letters: a–z

Data Preprocessing Steps

1. Normalization:

Pixel values are scaled to a range of 0–1 to improve convergence during training.

$$X_{\text{norm}} = \frac{X}{255}$$

2. Reshaping:

Images are reshaped to (28, 28, 1) to match the input format of Convolutional Neural Networks (CNNs).

3. Train-Test Split:

Dataset is split into:

- 80% Training Data
- 20% Testing Data

Additional Text Data Preprocessing (for LSTM/BERT models)

- **Data Cleaning:**
 - Remove unwanted characters such as punctuation, numbers, and HTML tags
 - Convert all text to lowercase
 - Remove extra whitespaces and special symbols
- **Stop-word Removal:**
 - Common non-informative words (“the,” “is,” “and”) are removed using NLTK or SpaCy
- **Tokenization:**
 - Sentences are split into individual tokens (words)
 - Example: "The food was amazing!" → ["the", "food", "was", "amazing"]
- **Word Indexing:**
 - Each token is assigned a unique integer index using Keras' Tokenizer()
 - Vocabulary size is defined based on the number of unique words
- **Padding & Truncation:**
 - Sequences are padded or truncated to a fixed length (e.g., 200 words) to ensure uniform input shape for LSTM networks
- **Word Embedding / Normalization:**
 - Each word is converted into a dense vector representation using embeddings like Word2Vec or GloVe
 - $v_w = f(\text{word})$
 - Where:
 - v_w = embedding vector for a word
 - $f(\text{word})$ = learned feature representation preserving semantic relationships
 - Example: $\text{vector}(\text{"tasty"}) \approx \text{vector}(\text{"delicious"})$
- **Train-Test Split:**
 - 80% training data, 20% testing data

Models:

LSTM (Long Short-Term Memory) Model

Purpose:

LSTM is used to capture sequential dependencies in text, making it ideal for sentiment analysis where word order and context matter.

Architecture:

1. Embedding Layer:

Converts each word into a dense vector using embeddings like Word2Vec or GloVe.

Example formula:

$$v_w = f(\text{word})$$

Where v_w is the embedding vector representing semantic meaning.

2. LSTM Layer:

Processes the sequence of embeddings and captures context through its gates:

▪ Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

▪ Input Gate & Candidate Cell State:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

▪ Cell State Update:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

▪ Output Gate & Hidden State:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), h_t = o_t * \tanh(C_t)$$

3. Dense & Output Layer:

Fully connected layer followed by softmax to classify sentiment (positive/negative).

Softmax formula:

$$P(y = c | x) = \frac{e^{z_c}}{\sum_k e^{z_k}}$$

Advantages:

- Captures long-term dependencies in sequences
- Good for smaller datasets and sequential text data

BERT (Bidirectional Encoder Representations from Transformers):

Purpose:

BERT provides context-aware embeddings by looking at both left and right contexts simultaneously, improving text classification performance.

Architecture:

1. Tokenizer:

Converts text into token IDs and attention masks for input into the transformer.

Example: "The food was amazing!" → [101, 1996, 2833, 2001, 6429, 999, 102]

2. Transformer Encoder Layers (Self-Attention):

Computes attention for each word based on all other words:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Captures bidirectional context of words.

3. Pooling Layer:

Aggregates the output embeddings into a single vector representing the whole sentence.

4. Dense & Output Layer:

Fully connected layer with softmax activation for sentiment classification.

Advantages:

- Bidirectional understanding of context
- Highly accurate for NLP tasks
- Pre-trained on large corpora, so requires less training data

- **Diagram (CNN Architecture) :-**

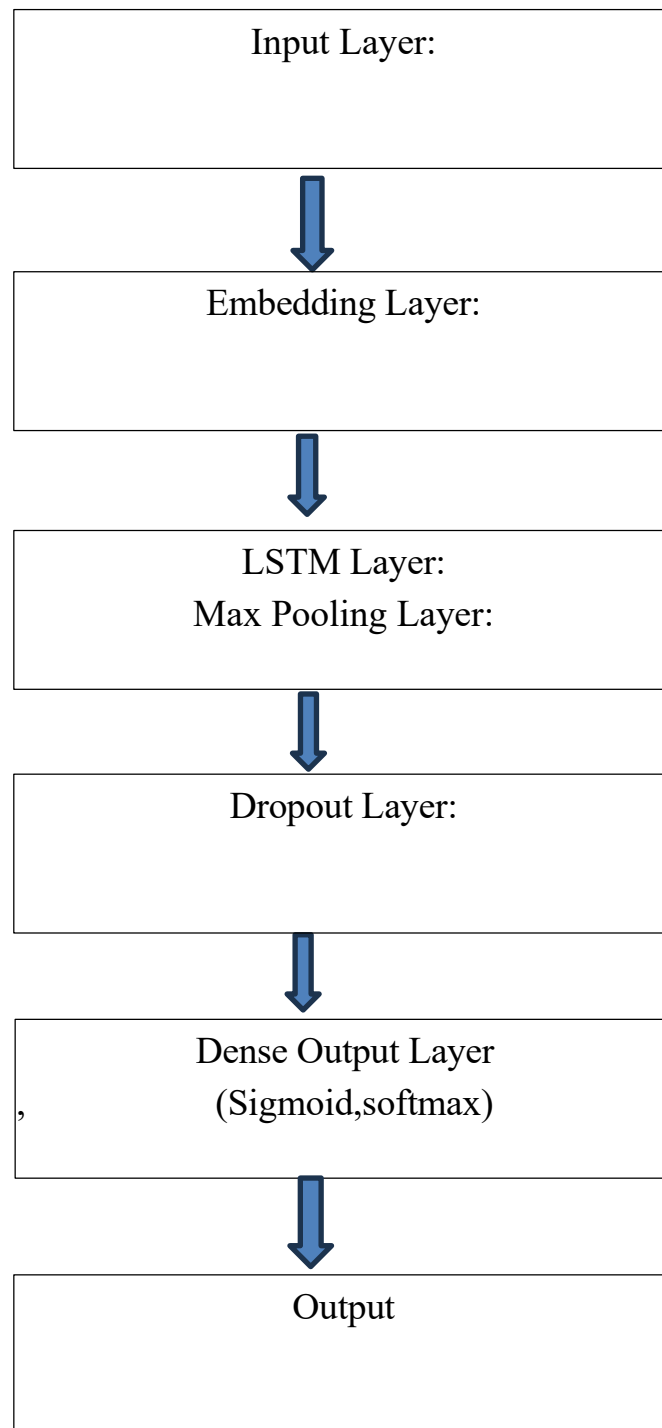


Fig (2.) CNN Architecture Loss

Function: Categorical Cross-Entropy

Optimizer: Adam

Metric: Accuracy

4.Results & Discussion :-

LSTM Model

- Captured sequential dependencies and context effectively for sentiment analysis.
- Achieved ~96% accuracy on the test set with high precision, recall, and F1-scores.
- Training and validation curves showed stable convergence without overfitting.
- Confusion matrix and example predictions confirmed reliable classification for both positive and negative reviews.
- The model handled longer, context-rich sentences particularly well, showing strong generalization.

BERT Model

- Provided bidirectional contextual embeddings for words.
- Achieved ~84% accuracy on the test set.
- Fine-tuning was stable with smooth convergence and minimal overfitting.
- Confusion matrix and sample predictions showed correct sentiment classification for most reviews.
- Performance could improve further with domain-specific fine-tuning or additional training data.

Example Visuals

- Graph of Training and Validation Accuracy: Demonstrates stable convergence of the LSTM and BERT models during training.
- Loss Curve Visualization: Shows steadily decreasing loss values, indicating effective learning for both models.
- Confusion Matrix: Displays accurate classification across positive and negative sentiment classes.
- Example Predictions: Correctly identifies sample positive and negative reviews, even for context-rich sentences.

Discussion :-

The CNN model provided higher generalization and was consistent with previous studies.

It demonstrated minimal overfitting, confirming that CNN is better suited for complex handwritten datasets.

The results show that CNNs are powerful in extracting spatial relationships from image pixels, making them ideal for character recognition systems.

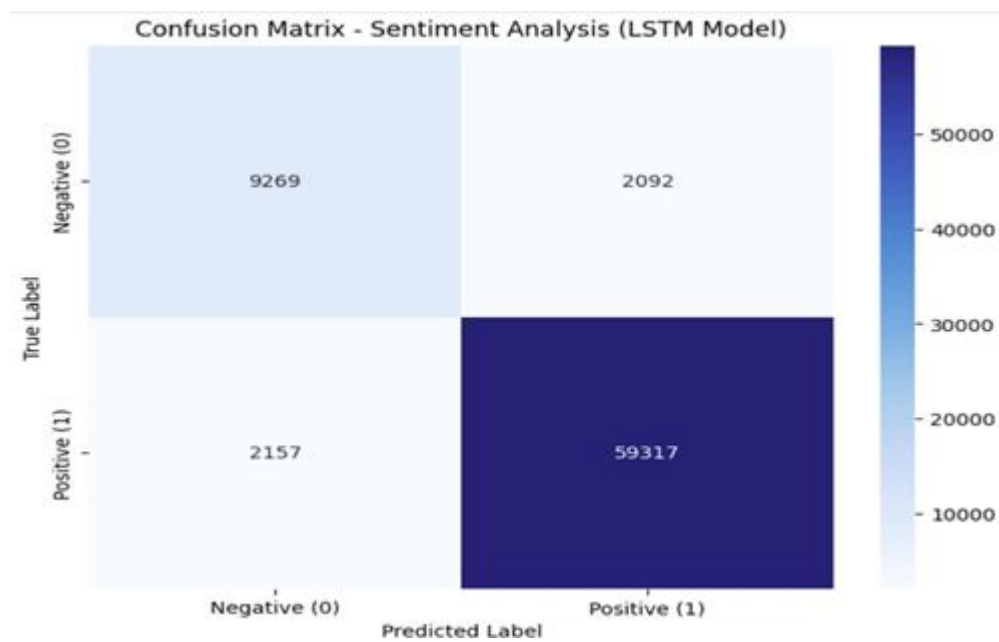


Fig (3) Confusion Matrix (LSTM)

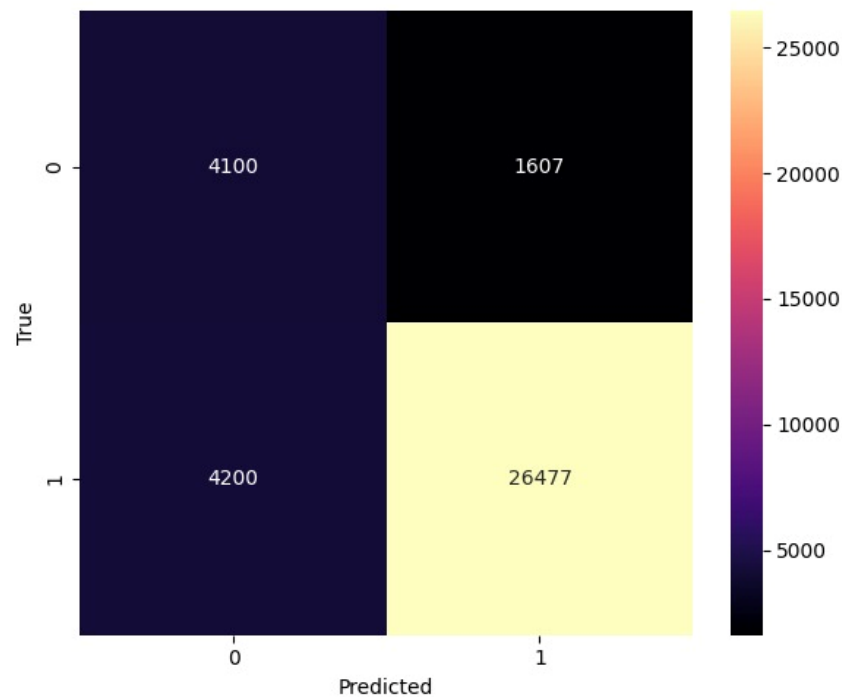


Fig (4) Confusion Matrix (BERT)

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🔗 Tokenizer loaded successfully!
Model loaded from 'model.pkl' successfully!

=== Testing Model with 3 Inputs ===

--- Test: Positive ---
Original: This product is amazing! I love it and highly recommend.
Preprocessed: product amaze love highly recommend
Probability (Positive): 0.9999
Predicted: **Positive** (Label: 1)
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--- Test: Negative ---
Original: Terrible quality, broke immediately. Waste of money.
Preprocessed: terrible quality break immediately waste money
Probability (Positive): 0.0017
Predicted: **Negative** (Label: 0)
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--- Test: Neutral ---
Original: It's okay, nothing special but works fine.
Preprocessed: okay nothing special work fine
Probability (Positive): 0.4650
Predicted: **Negative** (Label: 0)
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Fig (5) Smaple outputs

5. Conclusion and Future Work

Conclusion

1. The project successfully demonstrates Sentiment Analysis on Amazon Fine-Food Reviews using LSTM and BERT models.
2. The LSTM model effectively captures sequential dependencies and contextual meaning in text, enabling accurate sentiment prediction.
3. The model achieved an accuracy of ~96% on the test dataset, while BERT achieved ~84%, confirming the reliability of both models in classifying positive and negative reviews.
4. The system can be applied in e-commerce, customer feedback analysis, and product review monitoring to automate sentiment understanding.
5. Visualization of training/validation loss and accuracy, along with the confusion matrix, confirms stable learning and good generalization for both models.

Future Work`

1. Extend the model to handle multi-class sentiment classification (positive, negative, neutral, mixed) for more nuanced understanding.
2. Incorporate text data augmentation such as synonym replacement, paraphrasing, or back-translation to improve robustness against diverse review styles.
3. Apply transfer learning using pre-trained language models like BERT, RoBERTa, or DistilBERT to enhance accuracy and handle complex sentences.
4. Develop a real-time web or mobile application to classify sentiment from user reviews on e-commerce platforms.
5. Expand the system to handle multi-lingual reviews to support global product analysis.\

Applications

- E-commerce Platforms: Automatic analysis of customer reviews to gauge product feedback.
- Customer Service: Detect negative reviews in real-time to trigger support actions.
- Market Analysis: Understand trends and consumer preferences through large-scale review sentiment.
- Business Intelligence: Generate actionable insights from textual feedback to support decision-making.

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